CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: November 2019 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Name the three levels of Data Abstraction.**

Three Levels:

1. **Physical Level (Lowest):** Describes how data is stored on disk blocks.
2. **Logical / Conceptual Level (Middle):** Describes what data is stored and relationships among data.
3. **View / External Level (Highest):** Describes customized views for specific user groups.

Q2 (3M) **Differentiate between Entity Integrity and Referential Integrity.**

Differences:

- **Entity Integrity:** States that no primary key attribute value can be `NULL`, ensuring every tuple can be uniquely identified.
- **Referential Integrity:** States that a foreign key value must either match an existing primary key value in the parent relation or be `NULL`.

Q3 (3M) **Differentiate between Candidate Key and Primary Key.**

Explanation: A relation may possess multiple candidate keys (minimal super keys). The database designer selects one candidate key as the **Primary Key**, while all remaining candidate keys are designated as **Alternate Keys**.

Q4 (3M) **List any four disadvantages of File Processing Systems.**

Disadvantages: Data redundancy and inconsistency, lack of data independence, concurrent access anomalies, and poor security enforcement.

Q5 (3M) **Distinguish between Functional Dependency and Multivalued Dependency.**

Comparison:

- **Functional Dependency (X → Y):** For any two tuples with identical X values, their Y values must be identical (single-valued dependency). Basis of 2NF, 3NF, BCNF.
- **Multivalued Dependency (X ↠ Y):** The presence of (x, y_1, z_1) and (x, y_2, z_2) implies the presence of (x, y_1, z_2) and (x, y_2, z_1), representing independent multi-valued facts. Basis of 4NF.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain Second Normal Form (2NF) and Third Normal Form (3NF) with examples.**

2NF: In 1NF and no partial dependencies (every non-prime attribute depends on the whole candidate key).
3NF: In 2NF and no transitive dependencies (for every non-trivial FD X \to Y, X is a super key or Y is prime).

Q7 (7.5M) **Compare B-Tree and B+ Tree Index Structures.**

Comparison:

Feature	B-Tree	B+ Tree
Data Storage	Keys and data records/pointers stored in both internal and leaf nodes.	Data pointers stored **only in leaf nodes**; internal nodes store only routing keys.
Leaf Node Linking	Leaf nodes are not linked.	Leaf nodes are connected via a **doubly linked list**.
Range Queries	Slow (requires in-order tree traversal).	Extremely fast (scan linked list across leaf nodes).
Node Capacity	Fewer keys per node due to data pointers.	High fan-out, fewer levels, reduced disk I/O.

Q8 (7.5M) **Explain Conflict Serializability and Testing for Serializability using Precedence Graph.**

Precedence Graph Method:

1. Create a node for every committed transaction T_i.
2. Draw a directed edge T_i \to T_j if T_i performs a conflicting operation before T_j on the same data item (Write-Read, Read-Write, or Write-Write).
3. If the graph contains **NO cycle**, the schedule is conflict serializable. Topological sort gives the equivalent serial schedule.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the Three-Tier Client/Server Architecture of DBMS with Data Independence.**

Three-Tier Architecture:

- **Presentation Tier (User Interface):** Web browser or desktop client GUI (React, HTML).
- **Application Tier (Business Logic):** Web/Application server executing business validation and query processing (Java, Node.js, Python).
- **Database Tier (Data Storage):** Enterprise RDBMS managing database engine, query processing, transactions, and tables.

Provides enhanced security, scalability, and clean modular separation.

Q10 (15M) **Draw an ER Diagram for a Library Management System with full entities and relational schema.**

Entities: `BOOK`, `MEMBER`, `PUBLISHER`, `BORROWS`.

```
[MEMBER] ----- (M:N via BORROWS with IssueDate, ReturnDate) ----- [BOOK]
                                                                    | (N:1)
                                                                [PUBLISHER]
```

Q11 (15M) **Discuss SQL Characteristics and explain five Aggregate Functions with complete query examples.**

SQL Queries for Aggregate Functions:

```
-- 1. COUNT: Total employees in IT
SELECT COUNT(*) FROM Employee WHERE Dept = 'IT';

-- 2. SUM: Total payroll for Sales
SELECT SUM(Salary) FROM Employee WHERE Dept = 'Sales';

-- 3. AVG: Average salary in company
SELECT AVG(Salary) FROM Employee;

-- 4. MIN and MAX: Salary bounds
SELECT MIN(Salary) AS MinSalary, MAX(Salary) AS MaxSalary FROM Employee;
```

Q12 (15M) **Discuss Selection, Projection, and Join operators of Relational Algebra with detailed expressions.**

Detailed Operators:

- \sigma_{City='Noida'}(Customer) selects rows where City is Noida.
- \pi_{Name, Email}(Customer) projects distinct Name and Email attributes.
- Customer \bowtie_{Customer.ID = Order.CustID} Order performs equijoin between Customer and Order relations.

Q13 (15M) **Explain Database Recovery Techniques: Log-Based Recovery, Immediate Update, Deferred Update, and Checkpoints.**

Log-Based Recovery:

- **Deferred Update (NO-UNDO / REDO):** Updates are recorded only in the log until commit. During recovery, only committed transactions are REDONE; uncommitted transactions require NO UNDO.
- **Immediate Update (UNDO / REDO):** Updates can be written to disk before commit. Active uncommitted transactions at crash must be UNDONE; committed ones are REDONE.
- **Checkpoints:** Periodically flushes dirty log buffers to disk. Recovery scans backward only to the most recent checkpoint, dramatically shortening crash recovery time.
